Xibin Bayes Zhou (周禧彬)

Protein foundation models for search, dialogue, and design

Ph.D. Candidate in Computer Science, Westlake University & Zhejiang University, Hangzhou

zhoubay.github.io | zhouxibin@westlake.edu.cn | Google Scholar | ORCID | GitHub | LinkedIn

Education

- Ph.D. in Computer Science, Joint Program of Westlake University and Zhejiang University, Hangzhou, China (2022–2027, expected)
- B.Eng. in Data Science and Big Data Technology, Tongji University, Shanghai, China (2018–2022)
- High School, Tianjin Nankai School, Tianjin, China (2015–2018)

Research Interests

- **Structure-aware protein language modeling** — Integrating 3D structure tokens with sequence for general PLMs
- **Multimodal protein understanding** — Aligning sequence, structure, and language for protein search & annotation
- **AI-driven protein engineering** — Enzyme optimization, base editing, de novo design, and autonomous lab workflows
- **Democratizing PLM research** — No-code PLM training platforms and open collaboration for biologists

Honors & Awards

- **Oct 2025** — Best Poster Award, *Nature Conference on AI Augmented Biology* (Nanjing; poster on Evolla)

Community Impact

- Contributor to Hugging Face **Transformers** (Evolla integration) — commit
- **Evolla** post on X: 130K+ views — post
- **Evolla** on Hugging Face: ∼100K downloads/month — model
- **Pinal** featured in *Nature News* (May 2025) — article
- Research featured in **Westlake University** press (ProTrek / trimodal PLM) — news

Selected Publications

* equal contribution

- **Xibin Zhou**, *et al.* **Decoding the molecular language of proteins with Evolla**. *bioRxiv* (under review at *Nature*), 2025.
- F. Dai, *et al.*, **X. Zhou**. **Programmable design of functional proteins from natural language**. *Nature*, 2026. DOI: 10.1038/s41586-026-11205-z
- Y. He*, **X. Zhou***, *et al.* **PLM-assisted optimization of uracil-N-glycosylase enables programmable T-to-G/C base editing**. *Molecular Cell* 84(7), 1257–1270, 2024.
- H. Qian*, Y. Wang*, **X. Zhou***, *et al.* **ESM-Ezy: deep learning strategy for mining novel multicopper oxidases**. *Nature Communications* 16, 3274, 2025.
- J. Su*, Y. He*, S. You*, *et al.*, **X. Zhou**. **A trimodal protein language model enables advanced protein searches**. *Nature Biotechnology*, 2025.
- J. Su, *et al.*, **X. Zhou**. **Democratizing protein language model training, sharing and collaboration**. *Nature Biotechnology*, 2025.
- J. Su, *et al.*, **X. Zhou**. **SaProt: protein language modeling with structure-aware vocabulary**. *ICLR*, 2024.

Experience

- **Investment Intern**, ZhenFund — Jul–Sep 2023 (remote)

Extracurriculars

- Former **President**, Westlake Water Sports Club; **Rowing** (4 yrs, captain); **SUP** (3 yrs); sailing, dragon boat, and SUP polo

Languages

- Mandarin (native; 普通话水平测试一级乙等), English (proficient), Japanese (JLPT N2)
- Spanish, Latin (learning)